

EMCH 260

SOLID MECHANICS

Ming Hu

Associate Professor
Department of Mechanical Engineering
University of South Carolina
E-Mail: hu@sc.edu

Bending

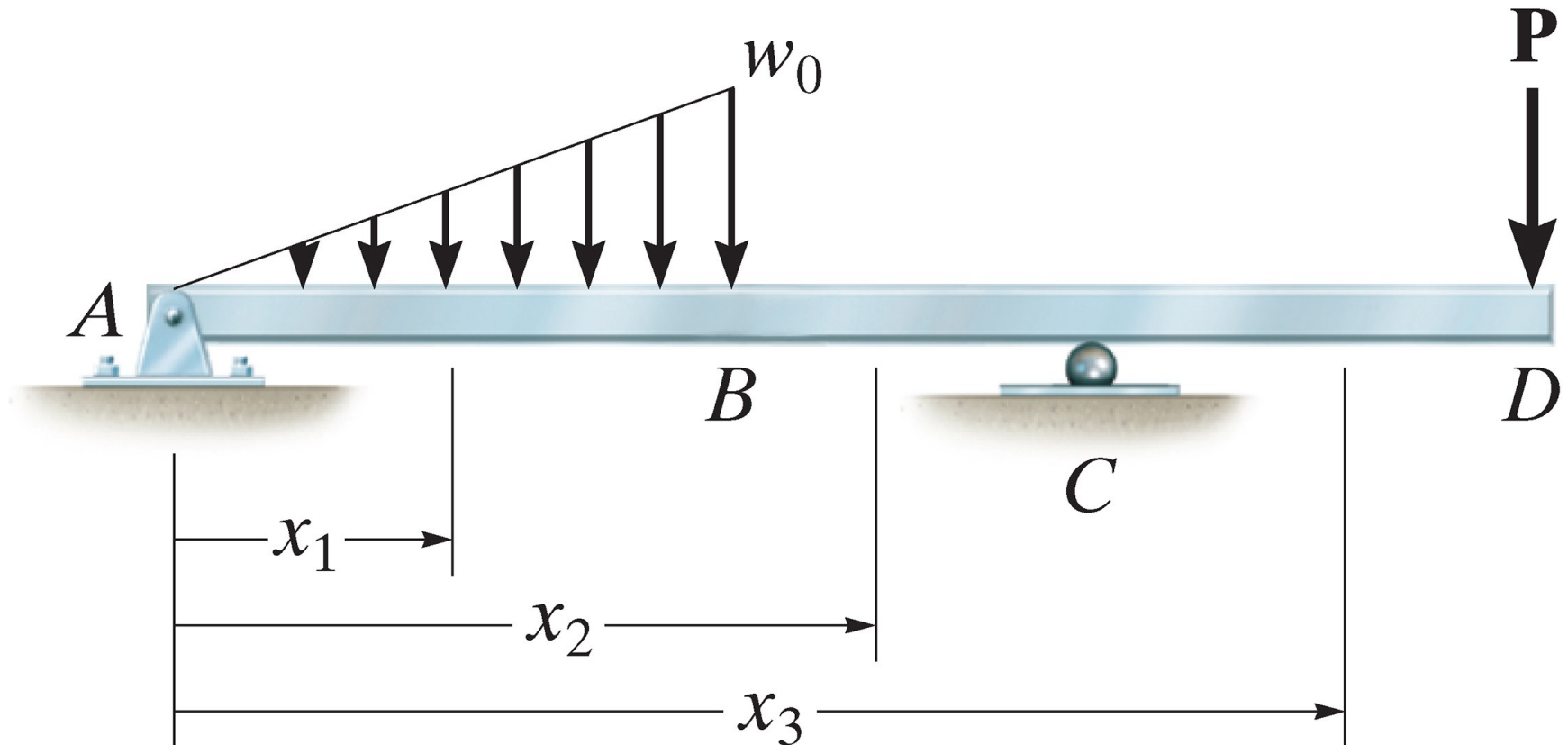
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Chapter 6.2

- ✓ Determine shear and moment diagrams by a simpler graphical method

PREVIOUS ON SHEAR AND BENDING MOMENT FUNCTIONS

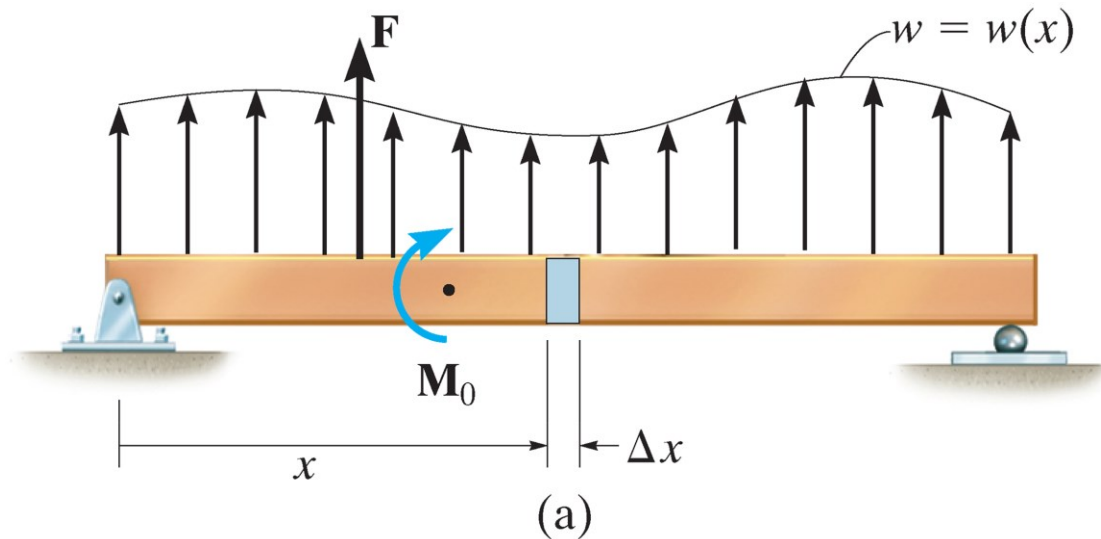
- Shear and bending-moment functions must be determined for each *region* of the beam between any two discontinuities of loading



Three different regions should be considered in this case.

CONTROL VOLUME METHOD OF ANALYSIS

Regions of Distributed Load

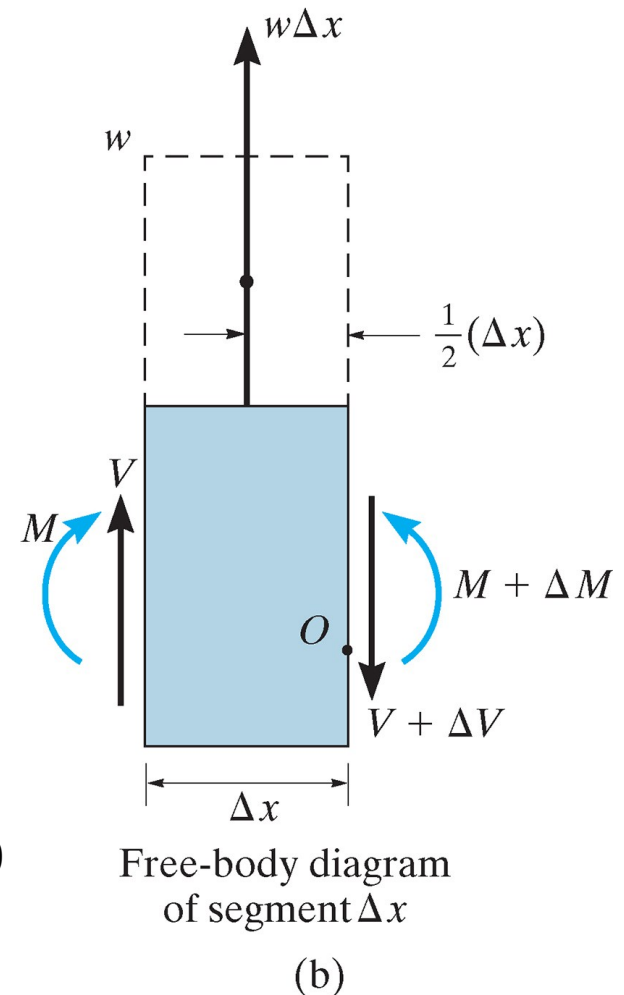


$$\sum F_y = 0 \quad V + w\Delta x - (V + \Delta V) = 0$$

$$\Delta V = w\Delta x$$

$$\sum M_O = 0 \quad -V\Delta x - M - w\Delta x \left[\frac{1}{2} \Delta x \right] + (M + \Delta M) = 0$$

$$\Delta M = V\Delta x + \frac{1}{2} w(\Delta x)^2$$



GRAPHICAL METHOD FOR CONSTRUCTING SHEAR AND MOMENT DIAGRAMS

Regions of Distributed Load

$$\frac{dV}{dx} = w$$

Slope of shear diagram at each point = distributed load intensity at each point

$$\frac{dM}{dx} = V$$

Slope of moment diagram at each point = shear at each point

IMPORTANT RELEVANT RELATIONSHIP

$$w \leftrightarrow V(x) \leftrightarrow M(x)$$

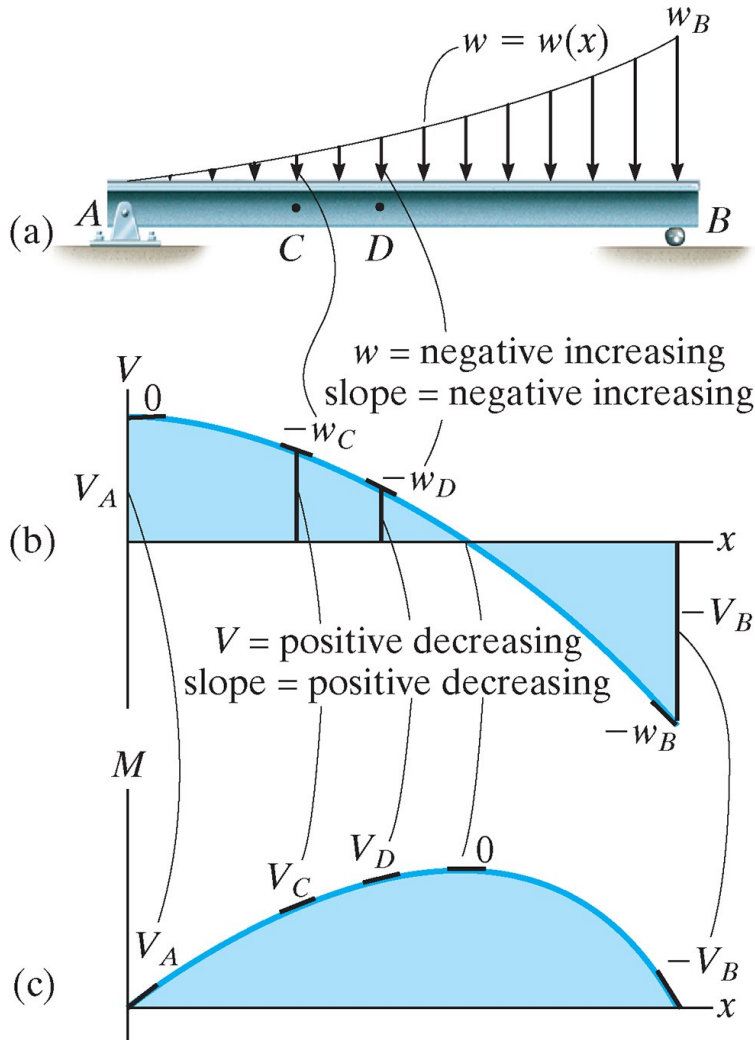
Integrate Form

$$\Delta V = \int w dx$$

Change in shear = area under distributed loading

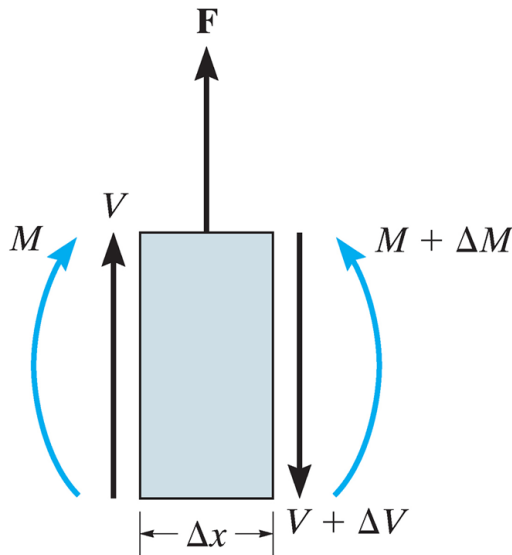
$$\Delta M = \int V dx$$

Change in moment = area under shear diagram



REGIONS OF CONCENTRATED LOAD

Regions of Concentrated Force and Moment

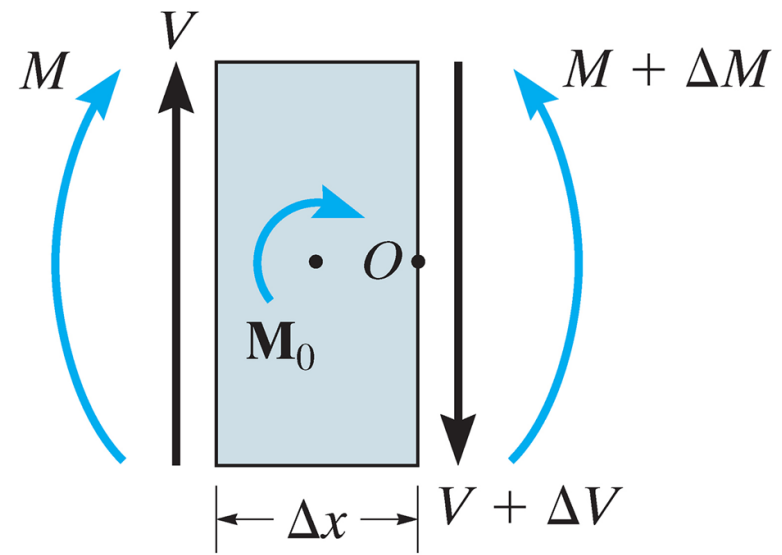


$$\sum F_y = 0$$

$$V + F - (V + \Delta V) = 0$$

$$\Delta V = F$$

Upward (downward) F results in positive (negative) change in shear (ΔV), corresponding to the “jump” upward (downward) in shear diagram



$$\sum M_O = 0$$

$$M + \Delta M - M_0 - V\Delta x - M = 0$$

$$\Delta x \rightarrow 0 \rightarrow \Delta M = M_0$$

Clockwise (counterclockwise) M results in positive (negative) change in moment (ΔM), corresponding to the “jump” upward (downward) in moment diagram

PROCEDURE FOR ANALYSIS

Step 1: Support Reactions

- Determine all reactive forces and couple moments acting on beam
- Resolve all forces into components acting perpendicular and parallel to beam's axis

Step 2: Shear Diagram

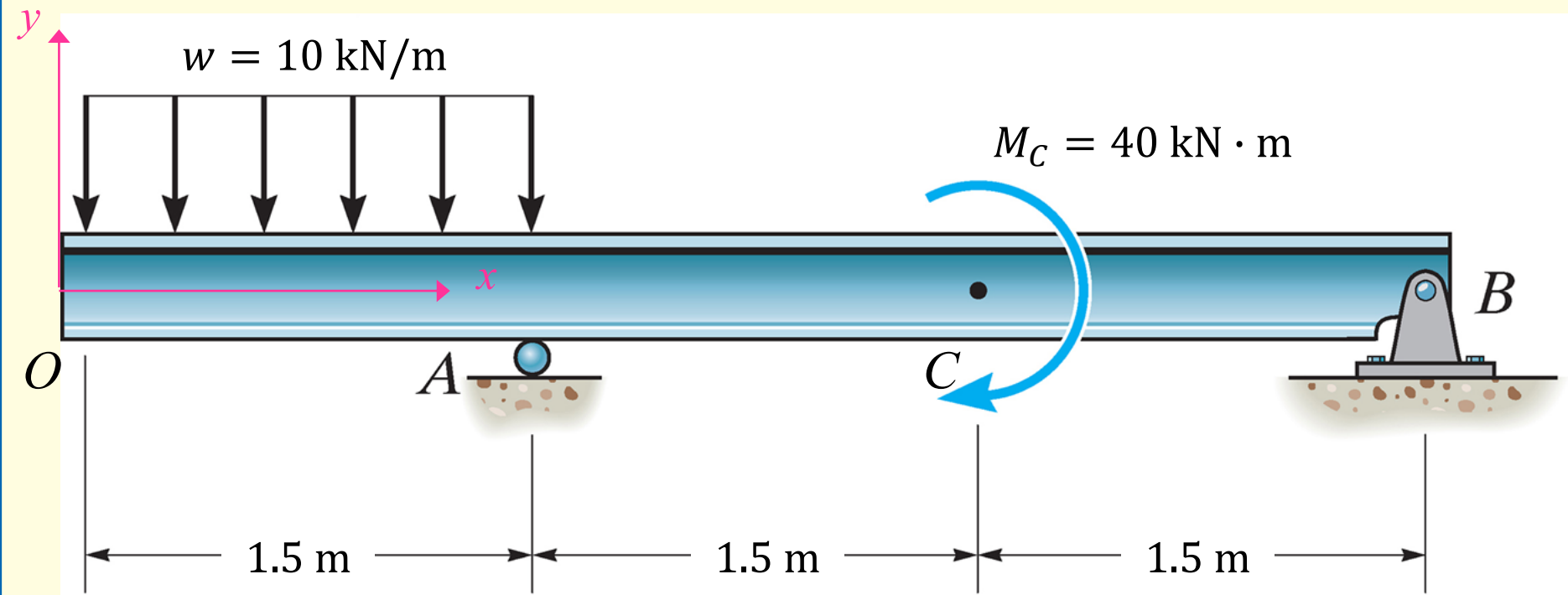
- Plot the known values of the shear V at the two ends of the beam
- Apply relationship between shear force and load $w \leftrightarrow V(x)$ for distributed force and $F \leftrightarrow \Delta V$ for concentrated force to determine the shear diagram

Step 3: Moment Diagram

- Plot the known values of the moment M at the two ends of the beam
- Apply relationship between moment and shear force $V(x) \leftrightarrow M(x)$ for anywhere without external moment, and $M_0 \leftrightarrow \Delta M$ for the point(s) with external moment to determine the moment diagram

IN-CLASS EXAMPLE

Using the graphical method, draw the shear and moment diagrams for the beam.



IN-CLASS EXAMPLE, cont'd

Solutions

Step 1: Support Reactions

Applying free-body diagram to the entire beam, the force equation of equilibrium gives:

$$\sum F_y = 0 \rightarrow F_A + F_B = 15000 \quad (1)$$

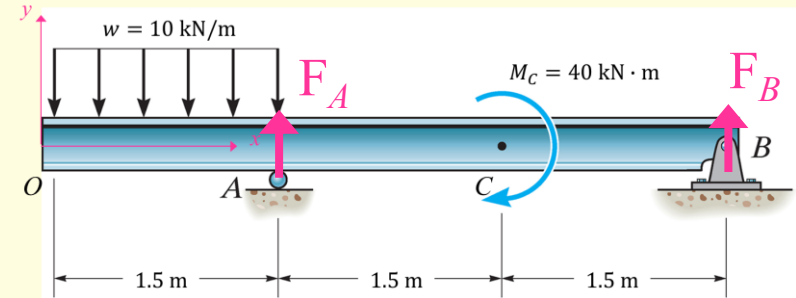
The moment equation of equilibrium with respect to point B gives:

$$\sum M_B = 0 \rightarrow F_A \times 3 + 40000 - 15000 \times 3.75 = 0 \quad (2)$$

Solving the above two equations gives:

$$F_A = 5416.7 \text{ N}$$

$$F_B = 9583.3 \text{ N}$$



IN-CLASS EXAMPLE, cont'd

Solutions

Step 2: Shear Diagram

a) Determine the shear at each end first (in particular at the origin of x-axis).

$$x = 0, \quad V = 0$$

$$x = 4.5, \quad V = -F_B = -9583.3 \text{ N}$$

b) Determine **V** diagram according to $\frac{dV}{dx} = w$ for OA region:

$$\frac{dV}{dx} = -10000$$

$$\rightarrow V = -10000x + C_1$$

The constant C_1 can be determined by applying the “boundary” condition ($x = 0, V = 0$). Thus, $C_1 = 0$ and $V = -10000x$ ($0 \leq x \leq 1.5$).

c) At point A, due to the support reaction force F_A , the **V** increases (upward jump) by F_A . See the shear diagram for details.

IN-CLASS EXAMPLE, cont'd

Solutions

Step 2: Shear Diagram (Cont'd)

d) For region $A \rightarrow C \rightarrow B$, there is no external forces and reaction forces, thus

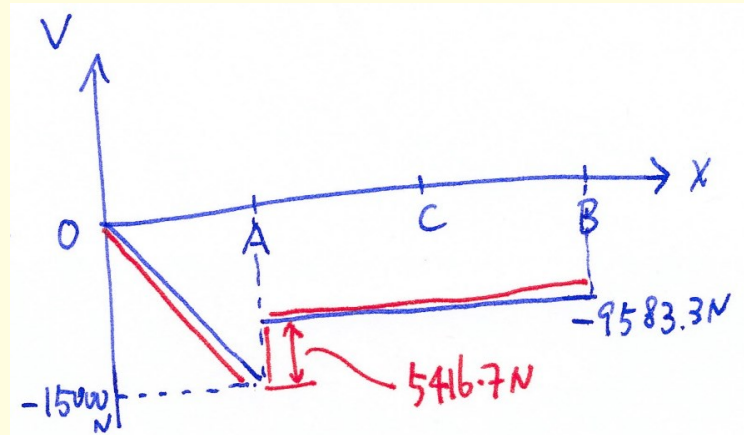
$$\frac{dV}{dx} = 0$$

$$\rightarrow V = C_2$$

The constant C_2 can be determined by the “**continuity**” condition at last point A after considering the upward jump.

$$\text{Thus, } C_2 = -15000 + 5416.7 = -9583.3 \text{ N.}$$

$$V = -9583.3 \text{ N} \quad (1.5 \leq x \leq 4.5)$$



VERIFY: The V for at end B is -9583.3 N, which is exactly equal to the value determined in a).

IN-CLASS EXAMPLE, cont'd

Solutions

Step 3: Moment Diagram

a) Determine **M** at each end first (in particular at the origin of x-axis).

$$\begin{aligned}x &= 0, & M &= 0 \\x &= 4.5, & M &= 0\end{aligned}$$

b) Determine **M** diagram according to $\frac{dM}{dx} = V$ for OA region:

$$\frac{dM}{dx} = V = -10000x$$

$$\rightarrow M = -5000x^2 + C_3$$

The constant C_3 can be determined by applying the “boundary” condition ($x = 0, M = 0$). Thus, $C_3 = 0$ and $M = -5000x^2$ ($0 \leq x \leq 1.5$).

IN-CLASS EXAMPLE, cont'd

Solutions

Step 3: Moment Diagram (Cont'd)

c) Determine **M** diagram according to $\frac{dM}{dx} = V$ for AC region:

$$\frac{dM}{dx} = V = -9583.3$$

$$\rightarrow M = -9583.3x + C_4$$

The constant C_4 can be determined by applying the “**continuity**” condition at point A ($x = 1.5$, $M_A = -5000 \times 1.5^2 = -11250 \text{ N} \cdot \text{m}$). Thus, $C_4 = 3124.95 \text{ N} \cdot \text{m}$ and **$M = -9583.3x + 3124.95$ ($1.5 \leq x \leq 3$)**.

d) At point C, due to the external moment, the internal resultant moment has upward jump of $40000 \text{ N} \cdot \text{m}$, i.e. the **M** jumps to $14375.05 \text{ N} \cdot \text{m}$. See the moment diagram for details.

IN-CLASS EXAMPLE, cont'd

Solutions

Step 3: Moment Diagram (Cont'd)

e) Determine **M** diagram according to $\frac{dM}{dx} = V$ for region CB:

$$\frac{dM}{dx} = V = -9583.3$$

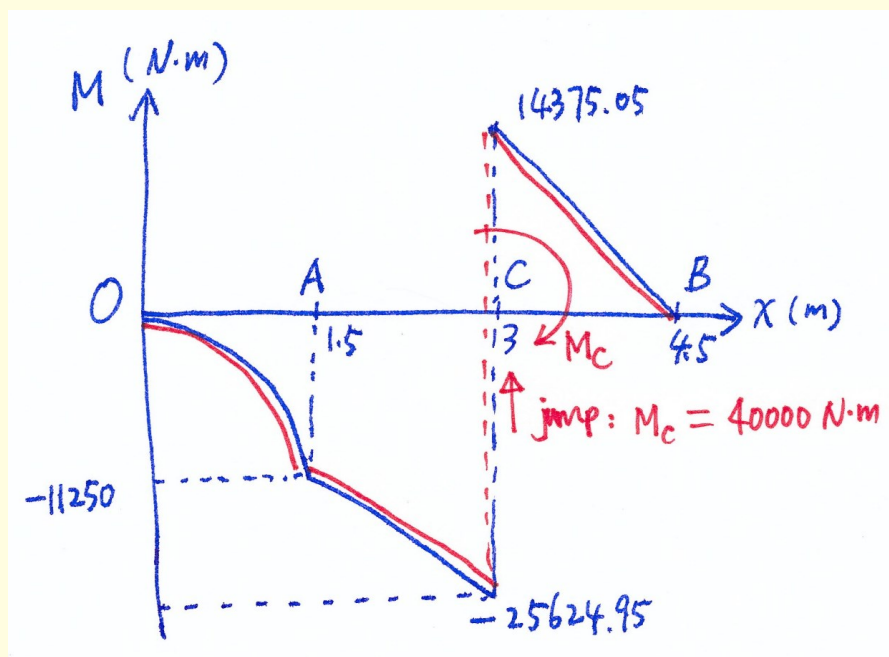
$$\rightarrow M = -9583.3x + C_5$$

The constant C_5 can be determined by applying the “**continuity**” condition at point C after considering the upward jump ($x = 3$, $M = 14375.05 \text{ N} \cdot \text{m}$).

Thus, $C_5 = 43124.95 \text{ N} \cdot \text{m}$

$$M = -9583.3x + 43124.95 \quad (3 \leq x \leq 4.5).$$

VERIFY: The **M** for at end **B** is zero, which is exactly equal to the value determined in a).



OUT-OF-CLASS EXERCISE

Solutions

You can also try to solve this problem by using traditional section method.
Try to do it by yourself and compare the results.

RELEVANT PRACTICE

- **Assignment 2 on Blackboard, problem #8 (go to Blackboard under the “Assignments” folder, **Assignment 2 due: 8:30 AM, Friday, Oct. 15th 2021**)**
- **Optional Extra Exercise 2 on Blackboard, problem #6 (go to Blackboard under the “Extra Exercises” folder, **Extra Exercise 2 due: 8:30 AM, Friday, Oct. 15th 2021**)**

OUT-OF-CLASS READING

- **Textbook, pp. 270 – 279**
 - **In particular, Examples 6.5 – 6.10 (pp. 274 – 279)**